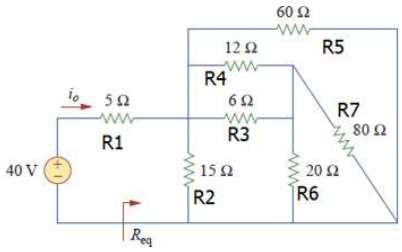
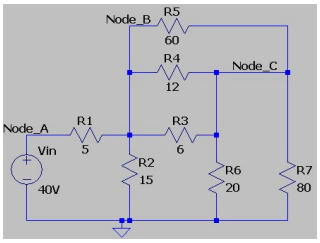
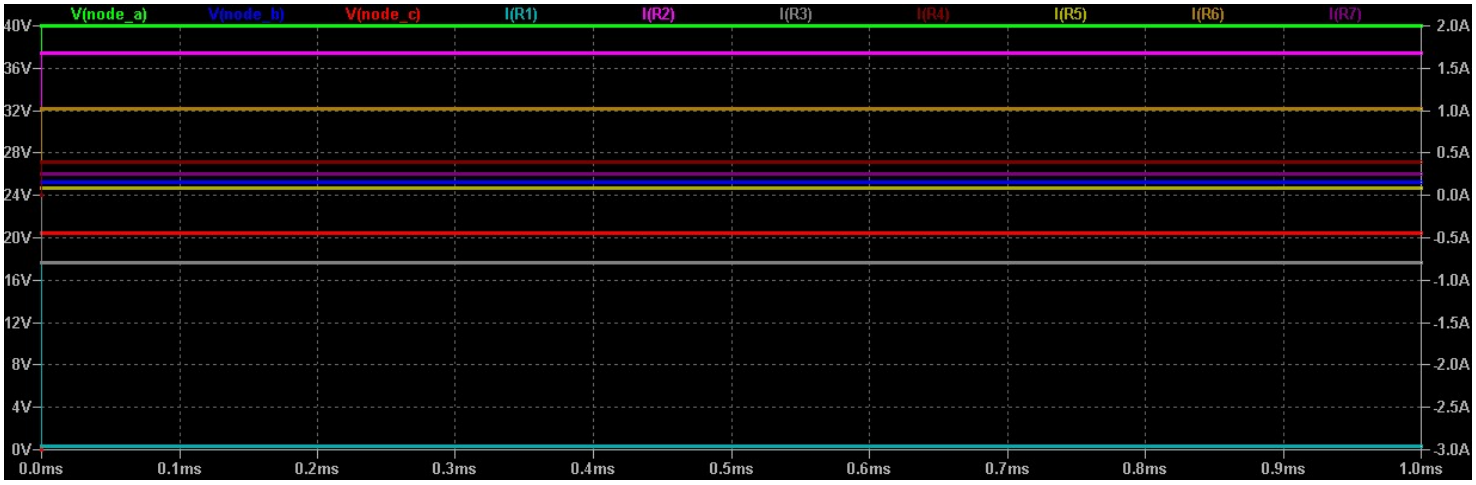
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3. Simulate the following circuit for 1ms, and write down the voltage at each node and the current through each resistor



a) LTspice simulation:





b) Demonstration

$R_1 := 5\ \text{ohm}$ $R_2 := 15\ \text{ohm}$ $R_3 := 6\ \text{ohm}$ $R_4 := 12\ \text{ohm}$ $R_5 := 60\ \text{ohm}$ $R_6 := 20\ \text{ohm}$ $R_7 := 80\ \text{ohm}$

$$R_{345} := \frac{1}{\left(\frac{1}{R_3} + \frac{1}{R_4} + \frac{1}{R_5}\right)} = 3.750\ \Omega \qquad R_{67} := \frac{1}{\left(\frac{1}{R_6} + \frac{1}{R_7}\right)} = 16.00\ \Omega \qquad R_{23456} := \frac{1}{\frac{1}{R_2} + \frac{1}{R_{345} + R_{67}}} = 8.525\ \Omega$$

$V_{a_gnd} := 40\ \text{volt}$

$$V_{b_gnd} := V_{a_gnd} \cdot \left(\frac{R_{23456}}{R_1 + R_{23456}}\right) = 25.21\ \text{V} \qquad V_{c_gnd} := V_{b_gnd} \cdot \left(\frac{R_{67}}{R_{345} + R_{67}}\right) = 20.43\ \text{V}$$

$$I_{R1} := \frac{V_{a_gnd} - V_{b_gnd}}{R_1} = 2.957\ \text{A} \qquad I_{R2} := \frac{V_{b_gnd}}{R_2} = 1.681\ \text{A} \qquad I_{R3} := \frac{V_{b_gnd} - V_{c_gnd}}{R_3} = 0.7979\ \text{A}$$

$$I_{R4} := \frac{V_{b_gnd} - V_{c_gnd}}{R_4} = 0.3989\ \text{A} \qquad I_{R5} := \frac{V_{b_gnd} - V_{c_gnd}}{R_5} = 0.0798\ \text{A} \qquad I_{R6} := \frac{V_{c_gnd}}{R_6} = 1.0213\ \text{A}$$

$$I_{R7} := \frac{V_{c_gnd}}{R_7} = 0.2553\ \text{A}$$

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